Pipeline:
Preparation:
1. Check cloud server: AutoDL (https://www.autodl.com/market/list)

2. Get more dataset from Jiahui (first check what previously Michel gave me)
→ problem: better to get one session with the same visual stimulation part ('polar' sessions)
→ All session files are found in: D:\Onedrive\ICPBR\forNikos\MatLab\DESNET\sesmonkeys
→ Check other E10 datasets and found:
→ 'polar'/'vspont': E10fV1, E10gb1, E10gH1, E10gt1, E10gW1,
→ 'estim': E10aW1, E10bf1, E10bv1,
→ 'vstim'/'vstim2': E10eA1

3. Organize data properly:
   a. original data: D:\DataPons
   b. processed data: D:\DataPons_processed
4. Think of proper organization of new files (https://chatgpt.com/c/69c21c13-a12c-8332-abf3-4266f8cc9a03)
→ ~~New folder: D:\Onedrive\ICPBR\Alberta\koopman_events\pipeline_v2~~
→Github repo: https://github.com/KaidiShao/LFP_BOLD_kpm_polished
5. After manual check of sessions/channels, keep a cfg function for each dataset in the folder 'configs'
→ for sessions: better to select polar and spont, but for those only with spont (e.g. F12m01), only with 'spont' sessions
→ **For channels, so far stay with both sr('hp') and 'pl', but later maybe try only 'pl'**

Pipeline:
LFP (before ResKoopNet):
0. **load data**:
→ function: \functions\io\load_blp_dataset.m
→ Load template: \scripts\templates\script_load_blp_data.m
1. Power spectrogram calculation
→ function: \functions\preprocessing\compute_blp_region_spectrograms.m
→ Script for each dataset: \scripts\templates\script_blp_calculate_spectrograms.m
→ Saving path: D:\DataPons_processed\f12m01\spectrograms\...
2. **Visualization of blp and spectrogram**:
→ function: plot_blp_segment_with_spectrogram.m
1. Spectrogram-based dictionary transformation
→ function: \functions\preprocessing\build_reskoopnet_dicts.m
→ Scripts: script_build_reskoopnet_dicts.m
1. ResKoopNet
→ Updated scripts: D:\Onedrive\ICPBR\Alberta\koopman_events\LFP_BOLD_kpm_polished\python_scripts\autodl
→ Original scripts: D:\Onedrive\ICPBR\Alberta\koopman_events\LFP_BOLD_kpm_polished\python_scripts\original
→ NOTE: used Codex to change solver_resdmd_batch2.py (autodl) to mixed precision to accelerate computation
→ Add email notification function: update after F12m01
→ 20260401: found that after 11 epoch, each train_psi takes much longer time (3000+s compared to the initial 1300+), optimized pipeline to solver_resdmd_batch3.py
1. Event detection pipeline (classical)
→ script: \LFP_BOLD_kpm_polished\scripts\script_compute_blp_event_density.m
→ function: "\LFP_BOLD_kpm_polished\functions\preprocessing\compute_blp_bandpass_events.m"
1. From event to density
→ function: "\LFP_BOLD_kpm_polished\functions\postprocessing\compute_blp_event_density.m"
→script: \LFP_BOLD_kpm_polished\scripts\script_compute_blp_event_density.m
1. Visualization of event detection:
→ Only LFP traces: script_plot_blp_segment_with_events.m
→ LFP traces + spectrograms: script_plot_blp_segment_with_spectrogram.m